

# Amitava Shee

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## Summary

Results-driven technology leader with 15+ years of experience specializing in building and scaling cloud-native recommendation and search engines at internet scale. Proven track record of delivering high-impact platforms that accelerate time-to-market and achieve significant cost savings. Experienced in leading multidisciplinary teams across AI, software engineering, data science, and platform engineering for global brands in music streaming, fintech, and healthcare.

## Experience

### Consultant, Services & Platform Engineering | SiriusXM | New York, NY | Nov 2022 - Present

- As acting Director of Personalization, engineered and launched a new recommendation engine for 30M consumers on a new streaming platform, achieving a market launch in a record 13 months.
- Optimized cloud costs by 68%, reducing monthly spend from \$1.57M to \$507K without impacting system performance.
- Led a 20-person engineering team to build and operate a machine learning platform for online and batch inference, processing over 1B records daily at scale.
- Architected a resilient, fault-tolerant platform that enabled rapid innovation and delivered new differentiated product offerings.
- Executing the delivery of MCP-connected apps for ChatGPT, Claude, and Gemini after successful proofs of concept, including AI/LLM and retrieval-augmented generation implementations using PostgreSQL, pgvector, and Neptune.

### Chief Technology Officer | reThought Insurance | Broomfield, CO | May 2021 - Mar 2022

- Led engineering teams across Software Development (9), Data Science (3), and DevOps (3) to build, operationalize, and support a new flood insurance underwriting platform.
- Designed and implemented a cloud-native platform on a self-hosted Kubernetes cluster, integrating with multiple third-party API services and risk analytics platforms.
- Enhanced risk scoring metrics by integrating flood exposure data from the National Flood Insurance Program (NFIP) and satellite data providers.

### Principal Engineer | AI & Machine Learning | Ithaca | Ann Arbor, MI | Jan 2017 - May 2021

- Drove product growth by implementing advanced search features, including a Learning to Boost (L2B) ML model to dynamically generate search engine parameters.
- Developed a continuous MLOps pipeline to data mine clickstream data, train models, and deploy them to production, accelerating the release cycle.
- Led the migration of distributed search from DataStax to Apache Solr on AWS EKS, resulting in significant cost savings and improved scalability at internet scale.
- Implemented a statistical A/B testing framework to validate product changes and guide data-driven decisions.

### Architect, Research Program | University of Michigan Medical School | Ann Arbor, MI | Nov 2013 - Jan 2017

- Delivered MedBus, a SOA-based data services platform that integrated large-scale health data for research, supporting a national healthcare research network.
- Secured approval for an API management solution by building a strong business case and helping bring IBM's products to the enterprise.
- Contributed to the president's \$100M Data Science Initiative and co-led a national committee that successfully delivered over 2M patient records.

### Enterprise Architect, Office of CIO | University of Michigan Health System | Ann Arbor, MI | Mar 2011 - Nov 2013

- Established an Enterprise Architecture program based on TOGAF, resulting in the development of a Healthcare Research IT Strategy and Roadmap.
- Evaluated and implemented the organization's first cloud vendor (AWS), paving the way for future cloud adoption.

### Information Architect, Enabling Technology | University of Michigan Medical School | Ann Arbor, MI | 2009 - 2011

- Led an applied research program (ENCORE) for medical education, developing rapid applications in Ruby on Rails to support educational research initiatives.

### Manager, Enterprise Architecture | University of Michigan Health System | Ann Arbor, MI | Jun 2007 - Jan 2009

- Led the development of clinical application architecture standards, successfully acquiring funding for capital projects through proposal creation and presentation.
- Served as a key stakeholder in the selection of the Epic healthcare system.

### Architect, Banking & Financial Services | Cognizant Technology Solutions | Indianapolis, IN | 2004 - 2007

- Partnered with Microsoft to implement a SharePoint-based consumer portal that integrates backend mainframe systems to present a unified view.
- Used BizTalk to integrate core banking systems, including PeopleSoft General Ledger and Loan Origination Systems.

**Software Architect** | CONTEXTWEB | New York, NY | 2002 - 2004

- Developed an Asset Backed Securities (ABS) online platform for the securities group at Royal Bank of Canada.

**Software Architect** | Techna International | Sunnyvale, CA | 1999 - 2002

- Acted as a consultant for IBM and Stanford School of Business, helping Galileo transition from mainframe to the web and implementing single sign-on (SSO) solutions.

**Senior Software Engineer** | CA Technologies | Kolkata, India | 1999 - 1999

- Developed a C++-based Microsoft COM/DCOM integration module in an object database product.

**Senior Software Engineer** | Tata Elxsi | Rockville, MD | 1995 - 1997

- Provided Electronic Data Interchange (EDI) consulting services for GE Information Services.

**Management Trainee** | Fujitsu ICIM | Pune, India | 1994 - 1995

- Developed system code in Unix and C/C++.

## Skills

**Engineering Leadership:** Led teams of 20+ engineers to deliver a new streaming platform in a record 13 months, enabling business innovation at scale.

**Data & AI Platforms:** Built and scaled recommendation and search engines, and led the development of MLOps pipelines for continuous model delivery.

**Cloud & Architecture:** Architected and modernized cloud-native platforms on AWS and Kubernetes, driving significant cost savings and operational efficiency.

**Product Innovation:** Drove product growth by implementing advanced search features and building applications with rapid turnaround times.

**Enterprise Strategy:** Established an Enterprise Architecture program and served as a key stakeholder in the selection of a new healthcare system.

## Education

Birla Institute of Technology | B.Eng. Computer Science

## Certifications

- MOR Leaders Program | 2016
- Scrum Alliance, License: 000339704 | 2016